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Paleomagnetism of the Permian–Triassic Siberian Traps Intrusions from the Kulumbe River Valley, Northwestern Siberian Platform

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Abstract—The Kulumbe River area occupies a transitional position in the Siberian Traps province due to being located in the junction zone of the structures of the Norilsk region and the Tunguska syncline. The Kulumbe river valley combines the characteristic volcanic formations of both regions; however, the sequence of the formation of the intrusive complexes and their correlation with the volcanics are still unclear. The paper presents new paleomagnetic data on the intrusive bodies of the Kulumbe river valley. It is shown that intrusive magmatism in the region continued for a long time spanning at least one interval of the reversed polarity and one interval of normal polarity. Based on the comparison of virtual geomagnetic poles, the Siluriyskaya intrusion (Norilsk complex) containing sulfide mineralization can be correlated to the Morongovskaya-Mokulaevskaya (Morongovsky-Mokulaevsky) formations and close to the formation time of the ore-bearing intrusions of the Norilsk region. The formation of the Kulyumbinskaya intrusion (Kureyskii (Kureysky) complex) and some other bodies of the Kureyskii and Kuzmovskii (Kuzmovsky) complexes is most likely to correspond to the Morongovskaya-Mokulaevskaya (Morongovsky-Mokulaevsky) time and also is close to the formation time of the ore-bearing intrusions. The reversely magnetized intrusive bodies in the Kulumbe River valley were formed after the completion of the main volcanic phase and are approximately synchronous with the Daldykan complex in the Norilsk region. The formation of the Katangskii (Katangsky) intrusions, just as in the central part of the Tunguska syncline, spans a long period of time and cannot be correlated to any particular volcanic formation. Finally, based on the detailed analysis of the paleomagnetic data we hypothesize that most of the intrusions of the Kulumbe River valley except for part of the reversely magnetized bodies have been formed before the end of the main regional deformations which roughly coincides with the termination of trap magmatism.

Keywords: paleomagnetism, Siberian Traps province, intrusions, Permian, Triassic
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INTRODUCTION

The Siberian Traps province is considered a reference example of the large igneous provinces (LIP) and is one of the largest regions of the Phanerozoic intraplate basaltic magmatism. The volcanic layers with a thickness up to 3.5 km (Fedorenko et al., 1996) and the related intrusive bodies fill the Tunguska syneclise in the Siberian platform and extend beyond it. The fact that, in contrast to many other similar provinces, the Siberian traps have been poorly affected by the subsequent deformations and have not been separated into parts due to the breakup of the continental blocks, makes them a promising object for studying large magmatic provinces and evolution of magmatism within these provinces, and for testing the hypotheses about the formation of giant regions of intraplate volcanism including the most widespread mantle plume concept (Dobretsov et al., 2010; Sobolev et al., 2011; Ernst, 2014). Besides, the Siberian Traps attract scientific interest due to the unique Ni-Cu-PGE deposits present in the Norilsk region and confined to the trap intrusions and the extensively discussed hypothesis of a probable causal relationship between the catastrophic magmatism and the global biotic crisis at the Permian–Triassic boundary (Wignall, 2001; Courtillot and Renne, 2003).

In the Siberian platform, several regions that differ in the compositional features and thickness of the magmatic products, tectonic structure, and metallogenic specialization (Fig. 1a) are distinguished within the province. These are the Norilsk, Maymecha–Kotuy, Putorana, Lower Tunguska regions and the Angara–Taseeva depression (e.g., (Zolotukhin et al., 1986; Fedorenko and Czamanske, 1997; Ivanov et al.,

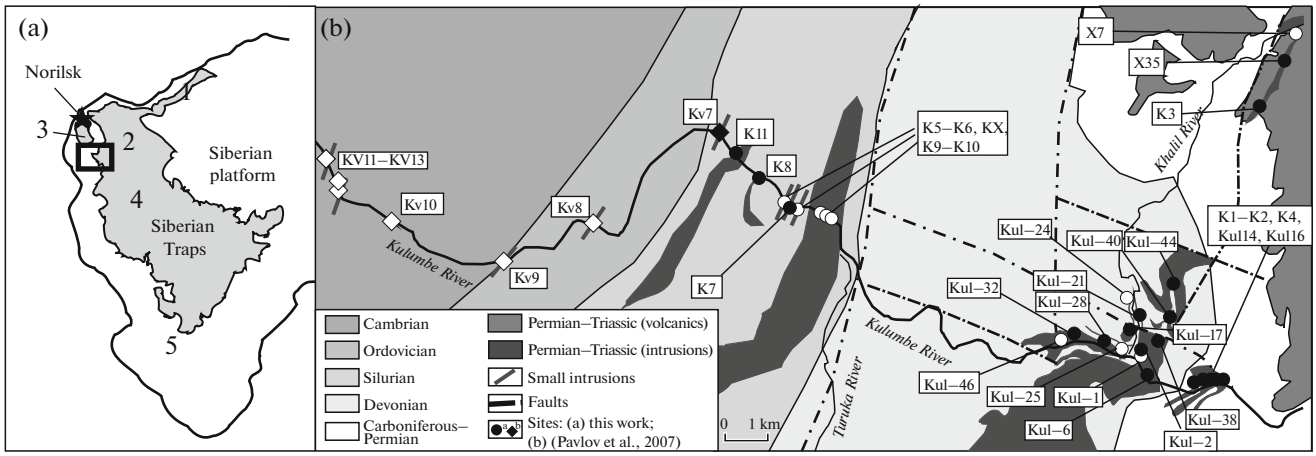


Fig. 1. (a) Position of study region in Siberian platform (numbers show regions of Traps province: 1, Maymecha–Kotuy; 2, Putorana; 3, Norilsk; 4, Nizhnyaya Tunguska; 5, Angara–Taseeva depression); (b) simplified schematic geological map of study region. Only sampled intrusions without differentiation into complexes are shown. Color for sampling sites shows magnetic polarity: normal by black, reversed by white.

2013)). Due to its exceptional economic importance, the Norilsk region has been studied most thoroughly. In the past three decades, extensive petrological and geochemical information was obtained on the volcanics and intrusive bodies of this region (Lightfoot et al., 1993; Wooden et al., 1993; Ryabov et al., 2001; Krivolutskaya, 2013). Rather large-scale studies associated with the specific composition of magmatites, namely, with the predominance of alkaline and ultrabasic rocks over tholeiitic basalts were also conducted in the Maymecha-Kotuy region (Fedorenko and Czamanske, 1997; Fedorenko et al., 2003). Other regions of the Siberian traps have been studied only fragmentarily.

In this work, we consider the Kulumbe River area whose nodal position in the structure of the province makes this region extremely interesting for the reconstruction of the Siberian Traps magmatism. On one hand, the Kulumbe River valley is located on the western margin of the Tunguska syncline where the most typical volcanic and intrusive formations of the "platform" stage of magmatic activity are developed (Al'mukhamedov et al., 2004). On the other hand, the geophysical data show that the Kulumbe region is superimposed on an offshoot of the buried Norilsk–Igarka Proterozoic paleorift system (Dolgal, 2012; Krivolutskaya et al., 2019) to which all the known commercial ore-grade massifs of the Norilsk region are confined. This key position of the Kulumbe river area is also reflected in the composition of the magmatic products: among the intrusive formations there are here the complexes that are characteristic of both the central part of the Tunguska syncline (Katangsky, Kuzmovsky) and the Norilsk region (Yergalakhskii (Yergalakhsky), Norilsk, Daldykan) and as well as the locally occurring (Kureysky) complex (Pavlov et al., 1988). South of the Kulumbe river valley there is the Dzhaltul ore-bearing massif. As far as it concerns stratified volcanic formations, the tuff-lava section is composed of the typical formations of the Norilsk region—from the Syverminskaya (Syverminsky) to the Nadezhdinskaya (Nadezhdinsky) ones (the bottom part of the section). The detailed material characterization of the Kulumbe intrusive bodies and volcanics was recently presented in (Krivolutskaya et al., 2020a; 2020b).

The position of the Kulumbe region at the junction of the two main regions of the Siberian Traps province and the presence of the objects prospective for the nickel–copper and platinum-group element mineralization in this region call for constructing a reliable scheme of the evolution of magmatic activity in the region and correlating the magmatic events to the volcanic units and intrusions in the neighboring regions. As of now, the substantiated data on the emplacement sequence of the intrusive complexes and the duration of magmatic events are absent.

Recent detailed studies demonstrated the efficiency of the paleomagnetic method in investigating the LIPs, in particular, in identifying the main magmatic episodes, estimating their duration and correlation with each other. Such data were obtained for the Siberian Traps (Gurevitch et al., 2004; Heunemann et al., 2004; Pavlov et al., 2011; 2019; Mikhaltsov et al., 2012; Latyshev et al., 2013; 2018) and for other LIPs in the world (Chenet et al., 2008; 2009; Moulin et al., 2012; Xu et al., 2018). The studies using paleomagnetic methods in the Siberian platform yielded a number of the important results: (1) based on the analysis of secular geomagnetic variations it was shown that the thickest volcanic sections in the Norilsk and Maymecha–Kotuy regions were formed in a series of short intense magmatic episodes (volcanic pulses) (Pavlov et al., 2011; 2019); (2) a new paleomagnetic pole NMK was calculated for the Permian–Triassic boundary for

the Siberian platform (Pavlov et al., 2019); (3) based on the comparison of the paleomagnetic directions it was shown that all the main ore-bearing Norilsk intrusions and their satellites were formed within a relatively short time interval synchronously with the Morongovskaya-Mokulaevskaya (Morongovsky-Mokulaevsky) formations of the Norilsk region (Latyshev et al., 2020a).

The currently existing paleomagnetic data for the Kulumbe region are relatively scarce and fragmentary (Pavlov et al., 2007; Latyshev et al., 2017). In this work, we intend to provide a detailed paleomagnetic characterization of the intrusions in the Kulumbe region and to use the obtained data for differentiating the intrusive complexes by age, correlating the main magmatic events to the neighboring regions of the Siberian Traps province, and estimating the duration of the magmatism.

STUDY OBJECTS AND GEOLOGICAL CHARACTERIZATION OF THE REGION

The study region spans the Kulumbe River's middle and lower reaches east of the Khantaiskoe reservoir, Krasnoyarsk krai. Geologically, the Kulumbe region is confined to the junction zone of the western slope of the Tunguska syncline and the Khantayka-Rybninskiy swell (Fig. 1b). The stratified sequence includes Cambrian–Devonian terrigenous-carbonate layers, Middle Carboniferous to Permian coal-bearing terrigenous formations pertaining to the Tunguska Group, and volcanics of the bottom part of the tuff-lava sequence (from the Syverminskaya (Syverminsky) to Nadezhdinskaya (Nadezhdinsky) formations) which are typically attributed to the Lower Triassic (e.g., (Fedorenko et al., 1996)). The sedimentary beds monoclinally gently dip east and so do the overlying volcanics composing the slope of the Nirungda trough. The Permian–Triassic intrusive bodies are, typically, sill-like dolerite and gabbro-dolerite intrusions with a thickness from 5–6 to 50 m, conformably intercalated among the Paleozoic sedimentary strata and, rarely, cutting across the bottom layers of the Syverminskaya (Syverminsky) Formation. We are not aware of the relevant isotopic age datings of the sampled intrusive bodies; however, the generally accepted opinion is that the intrusions in this region correspond to the Permian–Triassic Traps magmatism (e.g., (Pavlov et al., 2007; Krivolutskaya et al., 2020b)). The tectonic structure of the region is complicated by the regional Imangda-Letninskii fault zone which extends northeast and limits the Khantayka-Rybninskiy uplift.

In this work, we analyze paleomagnetic results from 36 sites of Traps intrusions including the new and previously published data (Pavlov et al., 2007; Latyshev et al., 2017). According to their geographical locations, the sampled intrusions can be divided into five localities (regions) briefly described below. Whether an intrusive body belongs to a particular complex or type was determined based on the field data, geological map (Pavlov et al., 1988) or geochemical characteristics published in (Krivolutskaya et al., 2020a; 2020b).

1. Two to three km upstream of the Khalil River mouth, five closely spaced sites (KUL-14, KUL-16, K1, K2, and K4) were sampled at the locality referred to as the Kulyumbinskaya intrusion. The field data are inconclusive of whether the studied outcrops belong to a single intrusive body; however, according to the geological map (Pavlov et al., 1988), the sampling area covers intrusions representing at least three complexes: the Yergalakhskii, Kureyskii, and Katangskii. Moreover, based on the geochemical indications, Krivolutskaya et al. (2020b) proposed to single out yet another, a new (Kulyumbinsky) complex in the nearby outcrops within this region. Thus, the Kulumbe intrusion is most likely to be a set of a few bodies that are spatially close but petrographically and geochemically contrasting. All the sampled intrusions are embedded into the Tunguska Group rocks and have a sill-like morphology.

2. Twelve sites of the intrusive bodies of the Katangskii complex (Table 1) and one body attributed to the Kureyskii complex, KUL-40, were sampled in the Khalil River mouth area. The sills are localized within the Devonian sediments and gently dip east conformably with the latter.

3. In the Khalil River valley, the Khalil intrusion was sampled at three sites (X-7, X-35, K3). Based on the geochemical data, the Khalil intrusion is related to the Kureyskii complex (Krivolutskaya et al., 2020a) and is subconcordantly intercalated in the lavas of the Syverminskii Formation.

4. Downstream of the Turuka river mouth in the Kulumbe valley, the intrusion (K7, Norilsk type of the Norilsk complex), Iltyk intrusion (K8, (Kruglogorsky) type of the Norilsk complex), and unnamed sills of the Daldykan (K5, K6, KKh, K9, K10), Katangskii (KV7), and Kuzmovskii complexes (K11) were sampled from a total of 15 sites. The intrusive bodies are localized in the Silurian sediments and gently dip east.

5. In the Cambrian–Ordovician sediments in the lower reaches of the Kulumbe River, thin intrusive bodies of the Daldykan complex were sampled at 6 sites KV8–KV13. The paleomagnetic results for these intrusions were previously presented in (Pavlov et al., 2007).

METHODS

The paleomagnetic samples were collected manually or using a mechanical Pomeroy rock coring drill. Their orientation was obtained with a magnetic compass under constant monitoring of the possible effect of highly magnetic rocks on the compass' needle. From ten to 25 oriented samples were collected from each site. The local magnetic declination was calcu-

Table 1. Paleomagnetic directions and VGPs calculated from sampled intrusions. *N*, number of samples (sites), Dg(s)/Ig(s), declination/inclination in geographic (stratigraphic) coordinate system; K, concentration; α95, confidence angle; Plat(g(s))/Plong(g(s)), VGP latitude/longitude in geographic (stratigraphic) coordinate system

Site	Complex/type	*N*	Dg, °	Ig, °	Ds,°	Is, °	K	α95°	Plong(g)°	Plat(g)°	Plong(s)°	Plat(s)°	A95°
Kulyumbinskaya intrusion													
KUL-14	Katangskii	8	338.5	81.3	58	83.3	34.5	9.6	44	81.3	127	71.4	18.6
KUL-16	Kureyskii	13	337.8	75.8	21.8	80.8	23.2	8.8	342.1	79.7	141	81.6	16.7
K1	Kureyskii	6	19.4	75.5	55.6	73.6	144.5	5.6	203.8	80.4	171.7	65.0	9.5
K2	Kureyskii	17	81.5	85.7	106.4	76.3	193.3	2.6	112.3	67.7	132.5	51.9	4.6
K4	Yergalakhskii	26	30.4	80.4	74.4	75.8	72.7	3.3	148.1	79.1	152.4	60.8	5.8
Mean direction and pole for Kulyumbinskaya intrusion		5	7.0	81.1	66.7	79.2	126.9	6.8	112.3	84.5	145.3	67.0	12.7
Khalil River mouth region													
KUL-2	Katangskii	10	148.8	65.3	133.9	59.0	32.7	8.6	112.7	27.7	126.4	23.2	11.1
KUL-6	Katangskii	14	106.7	75.2	96.7	50.6	30.8	7.3	133.9	50.3	160.7	26.4	8.1
KUL-17	Katangskii	13	39.2	67.9	53.6	60.6	9.2	14.5	201.5	64.5	194.5	51.4	19.3
KUL-21	Katangskii	14	119.2	45.6	115.2	36.7	20.2	9.1	143.0	10.0	148.8	10.0	8.1
KUL-28	Katangskii	8	57.0	66.0	65.9	57.2	31.8	10.0	185.1	55.9	184.9	43.6	12.5
KUL-32	Katangskii	6	48.4	56.9	56.5	48.9	52.0	9.4	203.1	49.6	199.0	39.8	10.1
KUL-38	Katangskii	13	100.4	71.9	96.9	62.0	146.3	3.4	142.4	47.7	154.4	37	4.6
KUL-40	Kureyskii	11	21.6	88.7	83.4	79.5	11.7	13.9	92.3	70.4	137.1	62.2	25.9
KUL-44	Katangskii	14	120.0	65.8	112	56.8	92.4	4.2	133.9	34.4	145.1	26.8	5.2
Mean direction and pole for normal polarity in situ		8	84.5	74.0			24.5	11.4	148.1	55.7			19.4
Mean direction and pole for normal polarity (stratigraphic coordinate system)		8			86.8	62.1	22.8	11.9			161.6	41.5	17.4
KUL-24	Katangskii	6	139.8	−70.1	168.5	−74.6	14.2	18.4	342.9	66.8	310.5	81.6	31.9
KUL-25	Katangskii	7	298	−65.3	290.6	−56.2	45.7	9.0	135.7	34.3	146.5	26.7	11.0
KUL-1	Katangskii	8	314.9	−68.5	301.8	−60.6	18.5	13.2	121.5	34.4	135.4	27.9	17.6
KUL-46	Katangskii	8	274.8	−64.1	273.6	−54.1	15.9	14.3	154.5	40.0	161.8	30.5	16.8

Table 1. (Contd.)

Site	Complex/type	*N*	Dg, °	Ig, °	Ds,°	Is, °	K	α95°	Plong(g)°	Plat(g)°	Plong(s)°	Plat(s)°	A95°
Mean direction and pole for reversed polarity		3	294.8	–66.8	287.9	–57.5	93.8	12.8	136.8	37.0	147.8	28.8	18.1
Khalil intrusion													
X-7	Kureyskii	8	289.9	–52.4	291.1	–42.5	21.0	12.4	148.8	23.5	151.1	15.3	12.0
X-35	Kureyskii	6	29.2	66.5	50.6	64.2	34.2	11.6	217.3	66.1	194.1	56.3	16.6
K3	Kureyskii	27	113.3	75.5	114.8	65.5	64.1	3.5	129.6	49.1	138.1	35.4	5.1
Turuka River mouth region													
KV7	Katangskii	6	31.2	88.3	112.9	70.8	63.0	8.5	94.5	70.8	134.8	42.7	13.8
K7	Norilsk	37	52.6	79.7	89.3	71.6	88.9	2.5	148.6	71.6	150.2	50.8	4.1
K8	Kruglogorskii	31	135.6	82.8	124.2	74.1	88.5	2.8	107.2	56.4	124.2	44.5	4.8
K11	Kuzmovskii	19	37.1	80.3	75.0	77.1	154.9	2.7	148.5	76.9	148.2	62.0	4.9
Mean direction and pole for normal polarity		4	62.7	84.4	101.7	74.2	153.3	4	119.8	70.5	138.1	50.5	13.4
K5	Daldykan?	56	283.2	–83.2	294.6	–74.7	201.9	1.3	117.7	61.9	129.4	47.6	2.3
KX	Daldykan?	17	266.9	–84.5	288.9	–75.1	132.5	3.1	116.9	66.1	132.4	49.6	5.4
K6	Daldykan?	35	266.1	–73.6	287.7	–57	111.9	2.3	149.2	54.3	148.2	28.4	2.9
K9	Daldykan	39	270.7	–75.3	284.1	–65.3	144.2	1.9	143.1	55.0	146.0	38.4	2.8
K10	Daldykan	19	311.8	–84.0	304.6	–72.1	102.3	3.3	106.4	58.9	125.7	41.6	5.5
Mean direction and pole for reversed polarity in situ		5	275.2	–80.4			188.3	5.6	128.2	60.3			10.2
Mean direction and pole for reversed polarity (stratigraphic coordinate system)		5			290.9	–69.0	98.2	7.8			137.1	41.5	10.9
Lower reaches of Kulumbe River (from (Pavlov et al., 2007))													
KV8	Daldykan	10	299.5	–82.1	306.6	–65.2	85	5.3	114.7	57.6	129.0	32.0	7.7
KV9	Daldykan	7	329.7	–81.2	323.6	–61.2	307	3.6	103.0	52.2	117.5	23.6	4.8
KV10	Daldykan	7	334.2	–80.5	313	–62.5	96	6.2	101.4	50.6	125.2	27.3	8.6
KV11	Daldykan	20	319.5	–82.3	308.1	–67.3	78	3.7	106.0	55.2	126.4	34.3	5.6

Table 1. (Contd.)

Site	Complex/type	*N*	Dg, °	Ig, °	Ds,°	Is, °	K	α95°	Plong(g)°	Plat(g)°	Plong(s)°	Plat(s)°	A95°
KV12	Daldykan	9	317.1	−77.6	306.2	−64.2	372	2.7	112.7	47.6	129.5	30.9	3.9
KV13	Daldykan	17	306.7	−74.7	309.1	−65.1	96	3.7	121.6	44.9	126.8	31.2	5.4
Mean direction and pole in situ		6	317.0	−79.9			468.2	3.1	110.1	51.6			5.7
Mean direction and pole (stratigraphic coordinate system)		6			311.4	−64.4	491.2	3.0			125.6	29.9	4.5
Mean direction and pole in situ from reversely magnetized intrusions in Turuka River mouth and lower reaches of Kulumbe River		11	298.7	−80.8			191.9	3.3	117.3	55.9			6.2
Mean direction and pole from reversely magnetized intrusions in Turuka River mouth and lower reaches of Kulumbe River (stratigraphic coordinate system)		11			303.1	−66.8	115.8	4.3			130.4	35.3	6.4
Mean direction and pole in situ from all sites of reversed polarity		14	297.1	−77.8			90.9	4.2	122.7	52.2			7.3
Mean direction and pole from all sites of reversed polarity (stratigraphic coordinate system)		14			299.0	−65.0	83.4	4.4			134.3	34.1	6.4
Mean direction and pole in situ from all sites of normal polarity		19	65.0	79.7			33.8	5.9	141.6	68.4			11.5
Mean direction and pole from all sites of normal polarity (stratigraphic coordinate system)		19			85.8	70.2	30.4	6.2					13.5
Mean direction and pole for normally magnetized sites of Katangskii complex (in situ)		5	120.4	74.5			48.5	11.1	126.3	47.1			19.5
Mean direction and pole for normally magnetized sites of Katangskii complex (stratigraphic coordinate system)		5			109.8	60.6	56.9	10.2			144.3	31.8	14.0

Mean pole for all sites calculated in geographic coordinate system: $N = 33$, Plong = 130.9°, Plat = 61.6°, A95 = 7.1°

Mean pole for all sites calculated in stratigraphic coordinate system: $N = 33$, Plong = 143.7°, Plat = 44.4°, A95 = 6.9°

Mean pole calculated in geographic coordinate system for sites with reversed polarity in Turuka River mouth region and in stratigraphic coordinate system for all other sites: $N = 33$, Plong = 143.1°, Plat = 47.4°, A95 = 7.1°

lated based on the IGRF model. The laboratory paleomagnetic studies and processing of the demagnetization results were conducted in the laboratory of the Main geomagnetic field and rock magnetism of Schmidt Institute of Physics of the Earth of the Russian Academy of Sciences (IPE RAS). All the samples were subjected to stepwise thermal demagnetization up to the complete removal of the magnetic remanence which took 10–16 steps and was in most cases achieved at temperatures of 520–600°C. Samples were demagnetized in non-magnetic ovens MMTD-80 with a residual field of at most 5–10 nT. The remanent magnetization of the samples was measured with JR-6 AGICO spinner magnetometer and 2G Enterprise cryogenic magnetometer. Several sister samples of the samples subjected to thermal demagnetization were demagnetized by the alternating magnetic field using a demagnetizing device built into a cryogenic magnetometer. The demagnetization by the alternating magnetic field (AF demagnetization) was carried out up to 130 mT. The measurements of remanent magnetization were processed using Enkin's software program (Enkin, 1994) with the magnetization components isolated by the principal component analysis (PCA) method (Kirschvink, 1980). The obtained data are analyzed using Fisher's statistics (Fisher, 1953).

The saturation magnetization versus temperature curves $Js(T)$ were recorded at heating to 700°C and subsequent cooling to room temperature on PMC MicroMag 3900 vibrating sample magnetometer in the magnetic field of 0.4 T. The temperature dependence of magnetic susceptibility æ(T) was measured on MFK-1FA kappabridge equipped with a CS-3 furnace apparatus for high-temperature measurements in the same temperature range. The hysteresis loops were recorded on PMC MicroMag 3900 vibrating sample magnetometer at room temperature with a saturation field of 1 T. The domain structure of ferromagnetic grains was assessed from the Day–Dunlop diagram (Day et al., 1977; Dunlop, 2002).

ROCK-MAGNETIC PROPERTIES

By the mineral composition of magnetization carriers, the studied intrusions of the Kulumbe river valley can be divided into three groups. Samples pertaining to group 1 are dominated by magnetite with the Curie points of 580–590°C as suggested by the temperature curves of saturation magnetization and magnetic susceptibility (Fig. 2a). The thermal curves of magnetic susceptibility also indicate the presence of hematite with Tc of ~675°C or a paramagnetic component (Fig. 2b). Most samples of this group demonstrate extremely stable behavior after heating. The samples of the Norilsk intrusions—the Iltyk (site K8) and Siluriyskaya ones (site K7)—and the intrusions in site K11 related to the Kuzmovskii complex are the examples of this group. The stability of this group of samples against heating indicates that hematite is likely to have been formed at high-temperature deuteric oxidation during cooling of a magmatic body and indirectly testifies to the primary origin of the remanent magnetization.

Group 2 differs from group 1 by the absence of hematite traces in the temperature curves of magnetic susceptibility and by the slightly lower Curie temperatures of the main magnetic phase—magnetite or Ti-poor titanomagnetite (Tc = 560–580°C) (Fig. 2c–2d). Samples of this group are also stable against heating; in some samples after heating, the Curie point in the cooling thermomagnetic curves æ(T) decreases to 500–550°C, which can probably be due to the homogenization of the decomposed titanomagnetite. This group includes the samples of the intrusions pertaining to the Daldykan complex as well as the Khalil intrusion (K3) and the Kulyumbinskaya intrusion (K2) pertaining to the Kureyskii complex. The thermal demagnetization data show that most of the samples in groups 1 and 2 contain a high-quality paleomagnetic signal.

Group 3 comprises samples from sites K1 and K4 which also refer to the Kulumbe intrusion. The thermal curves of saturation magnetization for the samples of this group have a characteristic concave shape typical of paramagnets (Fig. 2e). Despite low magnetic susceptibility in the samples, the susceptibility versus temperature measurements detected a small amount of magnetite with the Curie temperatures of 580–600°C. When a heated sample is cooled, the magnetic susceptibility sharply increases due to the new formation of magnetite (Fig. 2f). These magnetic properties of group 3 of the samples explain the relatively low quality of the paleomagnetic record at site K1 (see below).

Despite the difference in the thermomagnetic properties, the three groups of samples have fairly close hysteretic parameters falling in the pseudo-single-domain (PSD) region in the Day–Dunlop plot and varying within 0.08–0.27 (Mrs/Ms) and 1.4–3.6 (Hcr/Hc) (Fig. 2g). The cloud of the values for samples of groups 1 and 2 is elongated along the line of mixing of the single-domain (SD) and multi-domain (MD) grains (Dunlop, 2002) whereas group 3 of samples has higher Hcr/Hc ratios and shows a probable presence of superparamagnetic particles. The characteristic variations in the domain state of the samples comprising groups 1 and 2 are typical of Traps intrusions of the Siberian Platform (e.g., (Latyshev et al., 2020a)).

RESULTS OF PALEOMAGNETIC STUDIES

All the samples were subjected to a standard procedure of stepwise thermal demagnetization up to temperatures of 540–600°C, rarely up to 620°C. For many sites (K1–K11, Table 1), also mass demagnetization of the samples by the alternating magnetic field

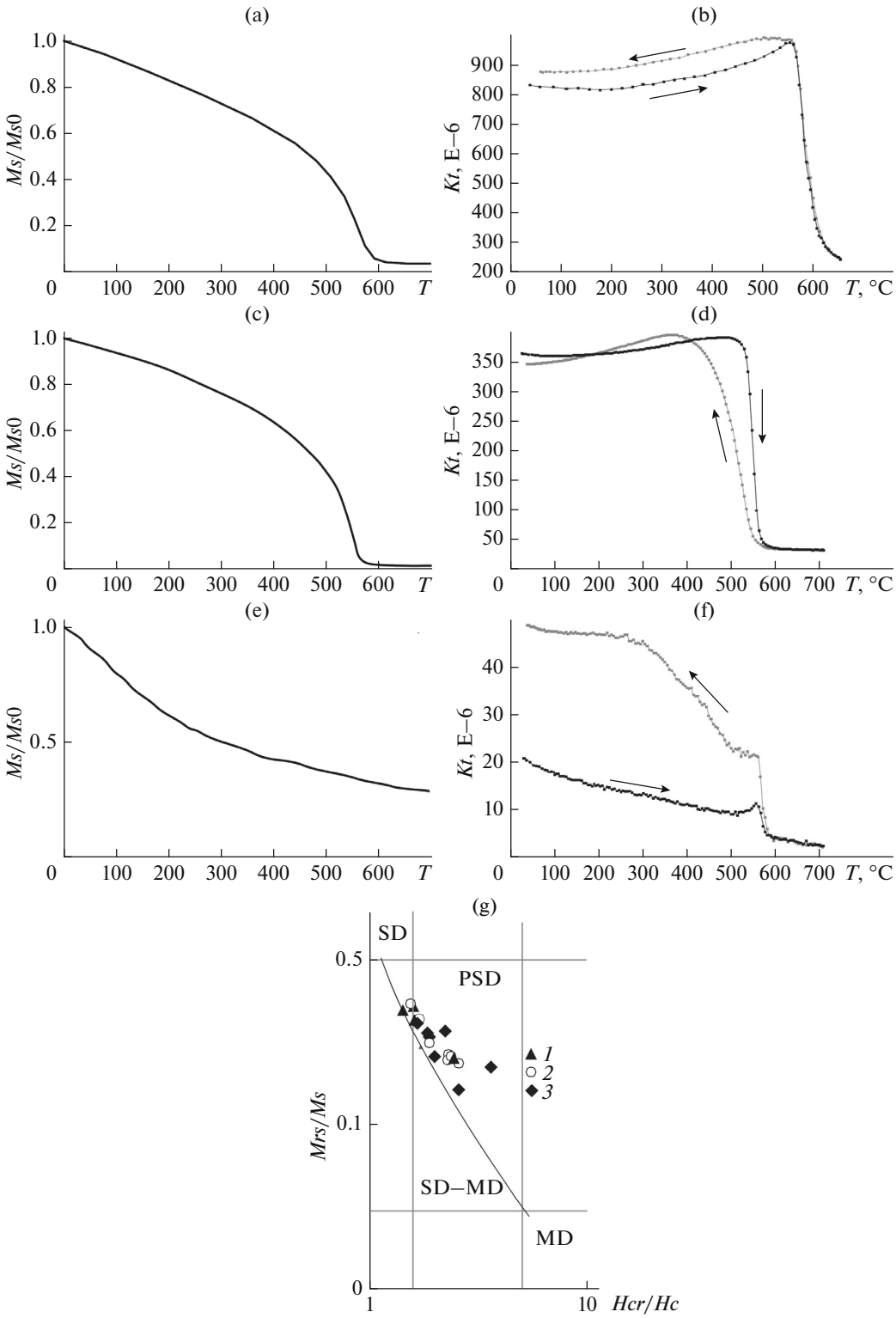


Fig. 2. Magnetic mineralogy of studied intrusive bodies. (a), (c), (e) Temperature curves of saturation magnetization (*T* is temperature (°C), *Ms* is saturation magnetization; *Ms*0 is saturation magnetization at room temperature): (a) sample K136, site K7, Siluriyskaya intrusion (Norilsk type); (c) sample K110, site K6, Daldykan complex; (e) sample K1, site K1, Kulyumbinskaya intrusion (Kureyskii complex). (b), (d), (f) Temperature curves of magnetic susceptibility (*K* is magnetic susceptibility; *T* is temperature. Arrows indicate heating/cooling): (b) sample K136, site K7, Siluriyskaya intrusion (Norilsk type); (d) sample K110, site K6, Daldykan complex; (e) sample K1, site K1, Kulyumbinskaya intrusion (Kureyskii complex). (g) Day–Dunlop diagram. *Ms* is saturation magnetization; *Mrs* is remanent saturation magnetization; *Hc* is coercive force; *Hcr* is remanent coercive force; SD is region of single-domain particles; PSD is region of pseudo-single-domain particles; MD is region of multi-domain particles; SD-MD is mixing line of single- and multi-domain grains (Dunlop, 2002); (*1*) Norilsk and Kuzmovskii complexes; (*2*) Daldykan complex; (*3*) Kureyskii complex.

(AF demagnetization) was carried out. For some sites (KUL-2, KUL-6, KUL-38, KUL-44), selective AF demagnetization was conducted for the samples whose thermal demagnetization yielded poor results. In the cases where the AF demagnetization proved to be more efficient, the mean paleomagnetic directions for these sites presented in (Latyshev et al., 2017) were recalculated with the allowance for the new data. Overall, in this work we discuss the results for 24 sites with new or recalculated data, for five sites whose results are presented in (Latyshev et al., 2017), and for seven sites whose results are described in (Pavlov et al., 2007) (Table 1).

Despite the fact that the paleomagnetic record in the studied intrusions varies from perfect to almost uninterpretable, the mean paleomagnetic directions were calculated for all the sites. For most objects, the calculated directions are close to those expected for the Permian–Triassic traps of the Siberian Platform (e.g., (Pavlov et al., 2019)); some exceptions are discussed below. Given that the sampled intrusions are located in the area confined to the (Imangda-Letninskiy) regional fault zone and may sit in the blocks with different time of fault deformations relative to the emplacement of the intrusions, in the first approximation we consider the paleomagnetic results for each of the studied regions.

The quality of the paleomagnetic signal in gabbro-dolerites of the *Kulyumbinskaya intrusion* varies from site to site. Two natural remanent magnetization (NRM) components are typically isolated in the samples. The low-temperature and low-coercive component is unblocked at 200–300°C and 5–10 mT and its directions are chaotic. The high-temperature normal-polarity remanence component is mainly demagnetized in the temperature interval 400–580°C (Fig. 3a), decaying to the origin in the Zijderveld diagrams (Zijderveld, 1967). The AF demagnetization isolates the same component with a median destructive field of 15–40 mT (Fig. 3b). However, in some samples from KUL-16 and KUL-14 sites, this remanence component has the untypical directions of traps, with anomalously low inclinations, whereas at the K1 site it is frequently not isolated because of the noisy signal. This is most likely to be due to the magnetic mineralogy peculiar of the rocks in the corresponding sites (see Section "Magnetic Properties"). This notwithstanding, we calculated mean paleomagnetic directions for all the sampled sites of the Kulyumbinskaya intrusion, and these directions proved to be located rather close to each other (Fig. 4a). Therefore, despite the fact that the sampled sites were attributed to different intrusive complexes, we may hypothesize that the rocks in these sites acquired remanent magnetization at a fairly close time.

Overall, gabbro-dolerites of the *Khalil intrusion* have a clear paleomagnetic signal. Besides the low-temperature component which is demagnetized at 250–300°C and has a significant scatter, most samples contain a single stable component unblocking in the temperature interval 450–580°C (Fig. 3c). In some samples, the paleomagnetic signal becomes chaotic at the highest temperatures (580–600°C) because of the formation of new magnetic minerals. The AF demagnetization of the samples from the K3 site isolated the same components; the median destructive field is, on average, 15 mT.

It should be noted that the high-temperature component at site X-7 has a reversed polarity whereas at the other two sites its polarity is normal. However, site X-7 is located virtually in the exocontact zone of a younger intrusion referred to the Katangskii complex and, therefore, it is should be taken into account that this site could be remagnetized by thermal or chemical processes during the emplacement of a more recent intrusion. The mean directions for sites X-35 and K3 which have normal polarity are located fairly far from each other in both the geographic and stratigraphic coordinate systems (Figs. 4a, 4b). In order to explain this phenomenon, we should either assume that the rocks at the sampled sites of the Khalil intrusion were magnetized at different times or that some unaccounted tectonic deformations are present there. The mean direction for site X-35 was calculated based on only six samples and has a fairly large confidence interval ($\alpha 95 = 11.6°$); besides, the samples from site X-35 have, on average, a substantially worse paleomagnetic signal that those from the K3 site. Therefore, in the subsequent calculations we only used data from the K3 site.

The quality of the paleomagnetic signal in the intrusions in the *vicinity of the Khalil river mouth* varies among different sites from good to extremely noisy. The low-temperature component is detected at all sites; it is removed at 250–350°C. The directions of this component have a significant scatter but in many cases are close to the present-day field in the sampling region; therefore, the low-temperature component is most likely to be viscous. The high-temperature component in most samples is isolated in the temperature interval 500–600°C. At nine sites, this component has normal polarity and at four sites it is reversed (Figs. 3d, 3e). At many sites, a number of samples also contain a medium-temperature component whose exact direction in many cases cannot be established because of the overlapping of its unblocking spectra with those of the low- and high-temperature components. Nevertheless, we have not revealed any regularity in the distribution of the medium-temperature component. The selective AF demagnetization has shown that the high-coercive component revealed by this method is close in direction to the high-temperature component.

The mean paleomagnetic directions for the high-temperature component were calculated for all 13 sites. Despite the rather large scatter of the directions ($\alpha 95 > 10°$ at 6 sites), they are mostly close to the

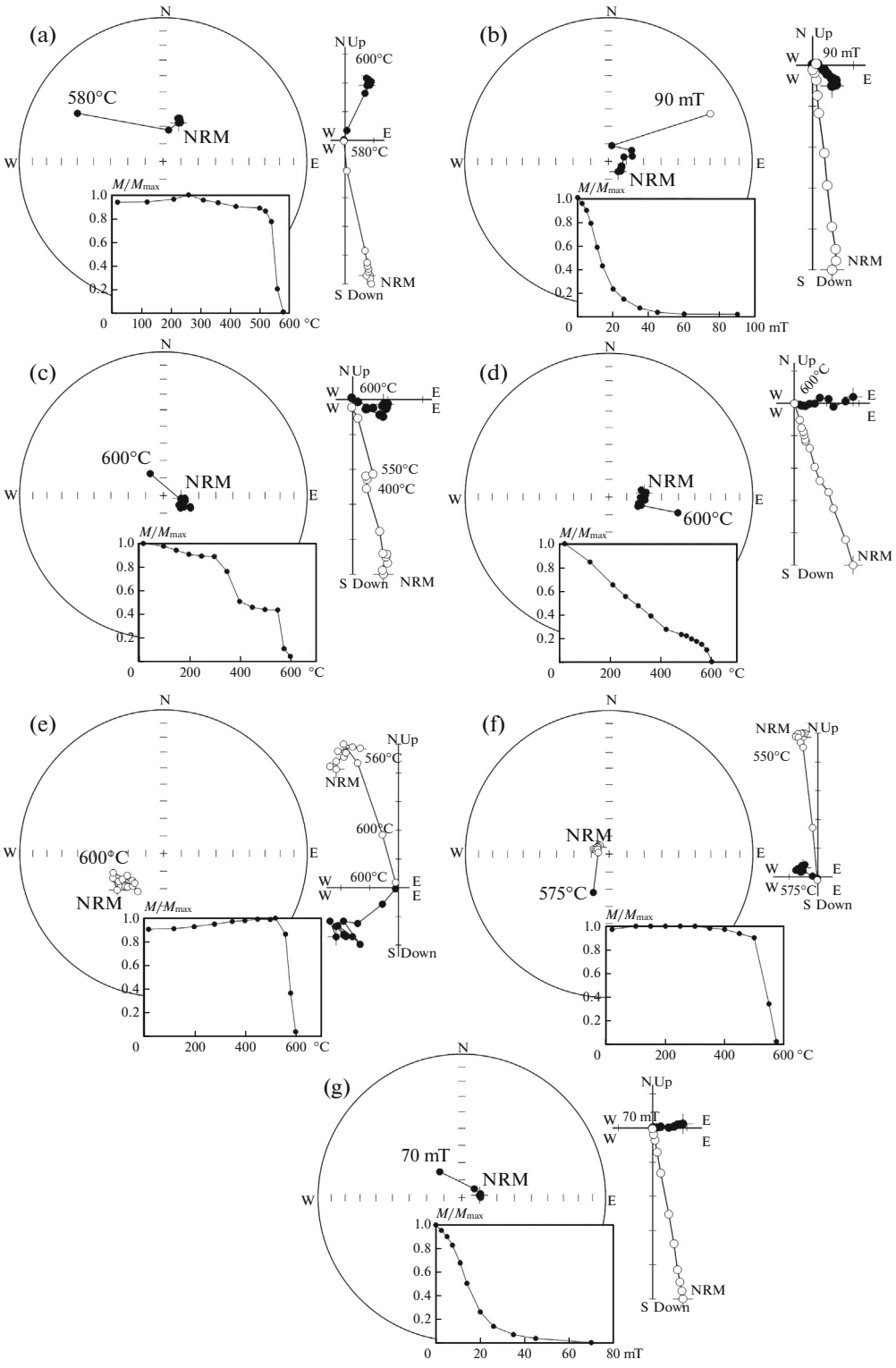


Fig. 3. Demagnetization results: (a) sample 68, site KUL-14, Kulyumbinskaya intrusion, thermal demagnetization; (b) sample K18, site K2, Kulyumbinskaya intrusion (Kureyskii complex), AF demagnetization; (c) sample 33, site K3, Khalil intrusion (Kureyskii complex), thermal demagnetization; (d) sample 95, site KUL-17, Katangskii complex, thermal demagnetization; (e) sample 244, site KUL-46, Katangskii complex, thermal demagnetization; (e) sample K89, site K5, Daldykan complex, thermal demagnetization; (g) sample 130, site K7, Siluriyskaya intrusion (Norilsk type), AF demagnetization.

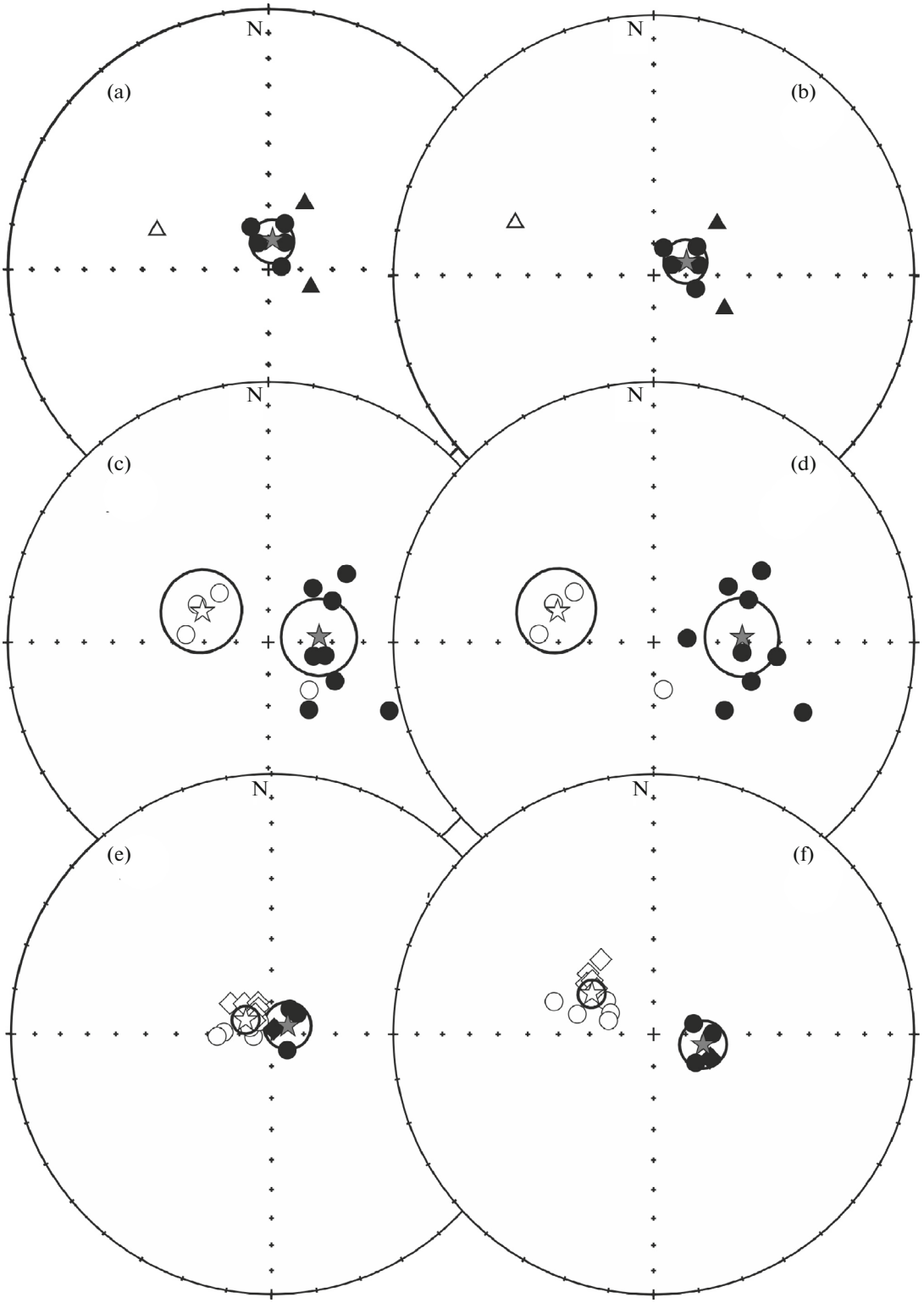


Fig. 4. Distribution of site-mean paleomagnetic directions. (a)–(b) Stereographic projection. Filled and open symbols correspond to lower and upper hemispheres, respectively. Mean directions are shown by asterisks with confidence intervals. Paleomagnetic directions of Kulumbinskaya and Khalil intrusions in (a) geographic and (b) stratigraphic coordinate systems. Sites of Kulumbinskaya and Khalil intrusions are shown by circles and triangles, respectively. Asterisk with confidence interval is mean for Kulumbinskaya intrusion. (c)–(d) Paleomagnetic directions of intrusions in Khalil river mouth area in (c) geographic and (d) stratigraphic coordinate systems. (e)–(f) Paleomagnetic directions of intrusions in lower reaches of Kulumbe river and mouth area of Turuka river in (e) geographic and (f) stratigraphic coordinates. Diamonds and circles correspond to sites from (Pavlov et al., 2007) and this work, respectively.

expected Traps directions (Figs. 4c, 4d). Sites KUL-21 with the anomalously low inclination for the Siberian Traps ($I = 45.6°$ and 36.7° in the geographic and stratigraphic coordinate system, respectively) and KUL-24 having the reversed polarity and southeastern declinations, also not typical for the Permian-Triassic traps, are the exceptions. Given the significant radius of the circle of confidence for the last site ($\alpha 95 = 18.4°$), we consider the results obtained for this site unreliable and do not use this direction in the subsequent calculations. A low inclination at the KUL-21 site may indicate its formation during a reversal or excursion of the geomagnetic field, which was previously established for the lava flows in the Norilsk region (Heunemann et al., 2004; Pavlov et al., 2011); however, we have no other evidence supporting this as of now.

The intrusions of the *Turuka River mouth region* typically contain a clear and easily interpretable paleomagnetic signal. The low-temperature and low-coercive component is present at all sites and is removed at 250–300°C, rarely at up to 400°C, or at 5–10 mT. The directions of this component are frequently close to the present-day directions and its origin is most likely to be viscous. The most stable magnetization component at five sites representing the intrusions of the Daldykan complex has a reversed polarity; at other sites of this region, this component is magnetized normally.

In the sills referred to the Daldykan complex, the stable component is isolated by thermal demagnetization in the temperature interval of 450–600°C and in the case of AF demagnetization it is completely unblocked at 50–90 mT (Fig. 3f). The median destructive field is 15–25 mT. The directions of the high-temperature high-coercive component are densely clustered within a site, whereas the site-means are located in the region expected for the reversely magnetized Traps although they significantly differ from the site directions in the lower reaches of the Kulumbe River (see below) (Figs. 4e, 4f).

The high-temperature normal polarity component was isolated at sites K7 (the Siluriyskaya intrusion, Norilsk type of the Norilsk complex), K8 (the Iltyk intrusion, Kruglogorskii type of the Norilsk complex), K11 (an intrusion pertaining to the Kuzmovskii complex) and at the spatially close site KV7 (Pavlov et al., 2007). At these sites, the temperature interval of this component ranges within 450–600°C. The AF demagnetization isolates this magnetization component in the fields of 10–70 mT, in some cases up to 100 mT (Fig. 3g). At site K11, thermal demagnetization in the interval of 150–500°C also isolates a medium-temperature component which is probably the result of remagnetization due to the emplacement of the nearby intrusions of reversed polarity. The site-mean directions of the most stable component are highly concentrated and have steep inclinations and east or northeast declinations which are typical for the Traps directions (Figs. 4e, 4f).

The results of the paleomagnetic investigation of the intrusions in the lower reaches of the Kulumbe River are presented in (Pavlov et al., 2007). A brief summary is that at all the five studied sites and in the baked exocontact zones of the sills, a stable reversed component is confidently identified. The site-mean directions form a tight cluster on the stereogram and are close to the other trap directions of reversed polarity (Figs. 4e, 4f).

Thus, for each of the sampled regions, fairly reliable paleomagnetic directions have been obtained in a number of cases. The primary origin of the remanent magnetization in the studied intrusions is supported by the following facts:

(1) the proximity of the bulk of the calculated directions to those expected for traps (e.g., (Pavlov et al., 2011));

(2) the positive contact test: the Early Paleozoic host sediments at some previously studied sites retain the primary pre-Traps remanence component (Pavlov and Gallet, 1998; Pavlov et al., 2008);

(3) the presence of the directions of normal and reversed polarity at close sites (the mouth regions of the Khalil and Turuka rivers) indicating the absence of the regional post-trap remagnetization;

(4) the stability of the magnetic properties and the absence of the magnetic mineralogical changes under heating for most samples.

Here, it should be stressed that based on the existing data, we cannot fully exclude the possibility of local remagnetization at separate sites during the emplacement of younger intrusions. Nevertheless, in this case, the age of the superimposed component will also correspond to one of the stages of the Permian–Triassic magmatism.

DISCUSSION

For further analysis, first of all, it should be ascertained whether the data need to be corrected for the inclination of the host strata, i.e., whether the intrusions were emplaced and acquired remanent magnetization before or after regional deformations. There is no conclusive answer to this question although the problem has been discussed before in connection with the traps of the Kulumbe River valley (Pavlov et al., 2007; Latyshev et al., 2017). One hypothesis states that the fold structures of the Khantaika–Rybninskiy swell were formed before the beginning of the Permian–Triassic magmatism in the Siberian platform (Kravtsov, 1967). In this case, the studied intrusions were intruded into the sedimentary sheets that were already gently inclined, and in the subsequent calculations we should use the in situ paleomagnetic directions. On the other hand, there is a widespread opinion that deformations in the study region developed synchronously with the folding in Taimyr (Late Triassic), i.e. after the end of trap magmatism (Malich,

1975; Nikishin et al., 2010). If so, when analyzing the paleomagnetic directions and poles, we should use stratigraphic coordinate system with the correction for the tilting of the sedimentary strata. Finally, yet another probable scenario is that both described scenarios took place, in some extent.

Let us firstly consider the distribution of mean directions for the sites having the reversed polarity except for sites KUL-24 and X-7 where the results are, in our opinion, not fully reliable (see above). The directions are somewhat more closely grouped in the geographic coordinates than in the stratigraphic coordinates ($Kg = 90.9$; $Ks = 83.4$, Table 1); the maximum concentration $K = 95.3$ is achieved at 40% unfolding (Watson and Enkin, 1993). At the same time, if we consider the results separately for each region, we obtain somewhat different results. For the intrusions around the Turuka river mouth, the in situ concentration is substantially higher than with the correction for the inclination of the country rock ($Kg = 188.3$; $Ks = 98.2$, Table 1), whereas in the other regions, the mean directions are somewhat more tightly grouped in the stratigraphic coordinate system. This fact indicates that the reversely magnetized intrusions in the Turuka river mouth area probably intruded after the regional deformations whereas the deformations in the other regions, at least partly, took place after the formation of the intrusions. In this case, the reversely magnetized bodies of the Turuka river mouth region correspond to the formation time of the Daldykan complex in the Norilsk region which also has the reversed polarity (Lind et al., 1994).

The similar analysis conducted for normal polarity directions also does not give a definitive result. If we consider the directions for all sites except for KUL-21 which has an anomalously low inclination (see above), the concentration parameter in the geographic coordinates is insignificantly higher than in the stratigraphic coordinates ($Kg = 33.8$, $Ks = 30.4$). The analysis of the data separately for different regions or for different complexes also does not provide a more definite result. Thus, we suppose a synfolding magnetization for most of the studied intrusions, which can be implemented in this case in two ways: (1) sills were intruded into the already partially inclined sheets and then underwent deformations; (2) some intrusions were formed before and some after the deformations. The bodies in the Turuka river mouth area referred to the Daldykan complex are the exception: they are most likely to have been emplaced at the post-folding stage. We note that if our hypothesis is correct, then the last significant deformations in the study region took place directly during the Permian–Triassic magmatism and were probably associated with the compensating subsidence within the Tunguska syncline due to the depletion of the intermediate or deep magma chambers.

We calculated the mean virtual geomagnetic pole from all sites except for X-7, KUL-21, and KUL-24 noted above in three versions: (1) in the geographic coordinate system for all sites; (2) in the stratigraphic coordinate system for all sites; (3) in geographic coordinates for the reversely magnetized sites in the Turuka river mouth and in stratigraphic coordinates for the other sites. The third calculation version yield the mean pole that is statistically indistinguishable from the NMK paleomagnetic pole (Pavlov et al., 2019) calculated from the volcanics of the Norilsk and Maymecha–Kotuy regions. The angular distance $\gamma = 5.2°$ is somewhat smaller than the critical angle $\gamma_{cr} = 6.3°$ calculated by the method proposed in (McFadden and McElhinny, 1990). We note that the mean poles calculated in only stratigraphic and only geographic coordinates substantially differ from the NMK pole (Fig. 5a). This result has two important implications. Firstly, it supports our hypothesis about relative ages of the magnetization and regional deformations presented above. Secondly, the coincidence of the mean pole with paleomagnetic indicates that the intrusions in the Kulumbe river area were formed for a fairly long time sufficient for the secular variation of the geomagnetic field to be averaged out. According to the modern estimates, this period is 10^4–10^5 years (Merrill et al., 1996; Tauxe et al., 2016).

Overall, the virtual geomagnetic poles (VGPs) calculated from the sites form a virtually isometric cloud of points on the stereogram (Fig. 5b). Although the density of the figurative points on the diagram is nonuniform, in the first approximation it is not possible to separate any representative dense clusters of the closely spaced VGPs which would correspond to the large-scale short magmatic events, in contrast to the other regions of the Siberian platform, e.g., the Angara-Taseeva depression (Latyshev et al., 2018). The group of the reversely magnetized intrusions in the lower reaches of the Kulumbe River (sites KV8 to KV13) is an exception. The diagram for these intrusions has an isolated cluster of points consisting of the VGPs of sites KUL-17, KUL-28, and KUL-32 (Katangskii complex) which substantially differs in longitude from the bulk of the poles and from the NMK paleomagnetic pole (by angular distance of ~30° in the latter case). This group of VGPs may probably testify to an excursion of the geomagnetic field or be the result of the undetected tectonic deformations and the related miscorrection for bedding attitude of the country rock.

In order to establish the place of the magmatic events that took place in Kulumbe river area in the evolution of trap magmatism of the Siberian Platform, we compared the VGPs from the studied intrusions with the mean poles from the volcanic formations of the Norilsk region (based on the data presented in (Pavlov et al., 2019)). In the case when the angular distance between the poles is large, we can state that the discussed magmatic events occurred at different time; when the poles coincide within the confidence inter-

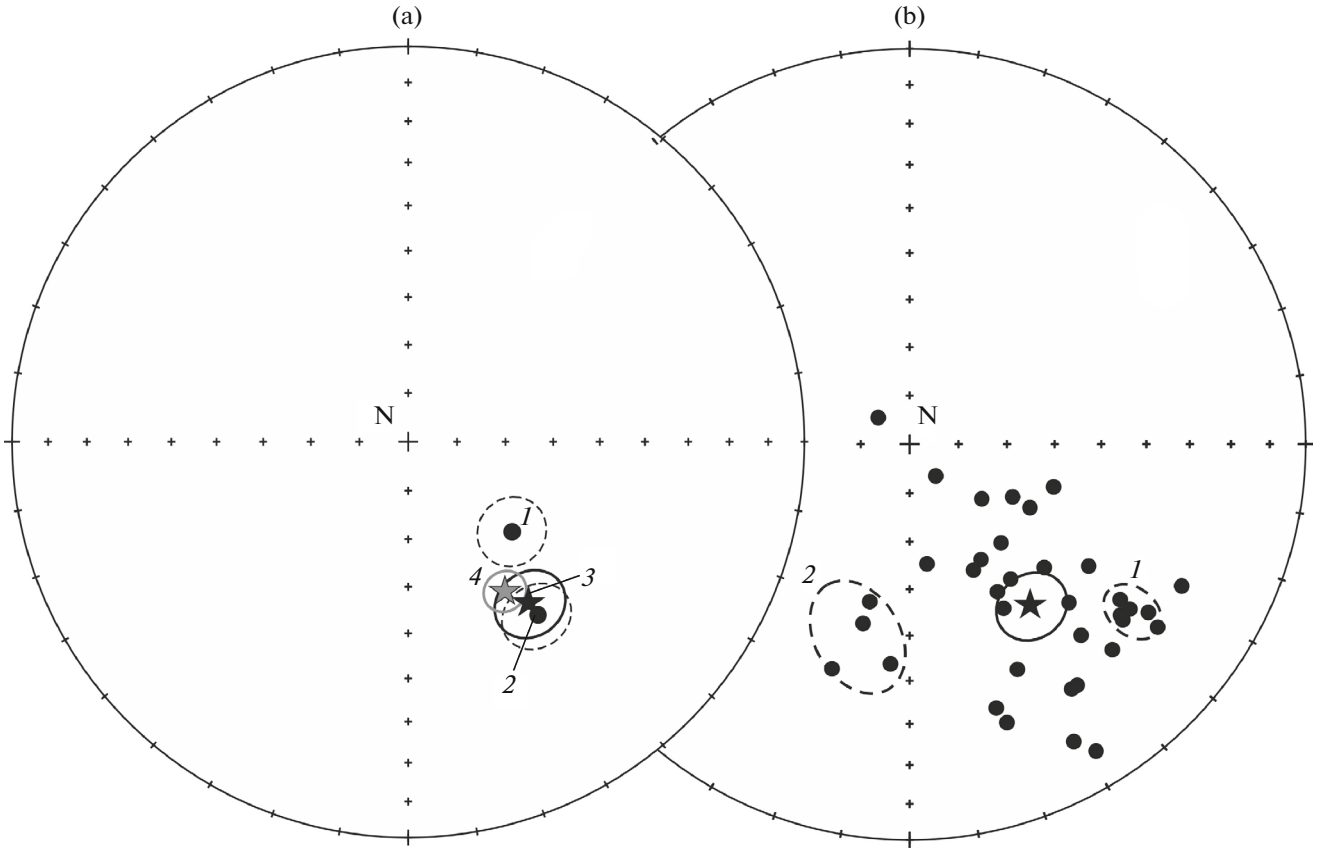


Fig. 5. Virtual geomagnetic poles calculated from paleomagnetic directions: (a) comparison of mean poles with Siberian Permian-Triassic paleomagnetic pole NMK (*1*, mean pole calculated in geographic coordinates; *2*, mean pole calculated in stratigraphic coordinates; *3*, mean pole calculated from reversely magnetized intrusions in Turuka River mouth in situ and from all other intrusions in stratigraphic coordinates; *4*, paleomagnetic pole NMK (Pavlov et al., 2019)); (b) distribution of site mean VGPs. Mean pole is shown by asterisk with confidence oval. *1*, group of poles from intrusions in lower reaches of Kulumbe River; *2*, "anomalous" VGPs (see text for explanations).

vals, the magmatic events are highly likely to be coeval. Previously, we used the similar technique for correlating the intrusions of the Norilsk region with the volcanics (Latyshev et al., 2020a). Given that in the volcanic section of the Norilsk region we have previously identified the volcanic pulses—the successive series of the flows with very close paleomagnetic directions, in this study we calculated the reference poles for the formations in two versions: (1) the flow-mean poles (hereinafter version I); and (2) the mean poles over pulses and over individual directions (hereinafter version II) (Table 2). As will be shown below, these two calculation versions lead to the interpretations that are somewhat different albeit not fundamentally.

Given that the VGP distribution (Fig. 5b) barely contains dense clusters which could correspond to the intense magmatic events, below, to compare the poles of the intrusions with those of the volcanic formations, we consider several intrusion groups combining the intrusive bodies that pertain to the same complex and are close geographically or by the paleomagnetic characteristics: (1) the intrusions of the Katangskii complex mainly present in the Khalil river mouth area; (2) the spatially close sites of the Kulyumbinskaya intrusion which have close paleomagnetic directions; (3) the reversely magnetized intrusions. Some intrusions which are the only sampled members of a particular complex or type, e.g., the Siluriyskaya intrusion or the Khalil intrusion, were considered individually. We note, however, that in the latter case, the correlation is, in our opinion, less reliable.

Among the studied intrusive bodies of the Kulumbe valley, the Siluriyskaya intrusion (site K7, Norilsk type) where several studies detected sulfide mineralization is most interesting. In the geographic coordinates, the VGP calculated for this intrusion is fairly distant from the poles of all the volcanic formations of the Norilsk region (Figs. 6a, 6b) whereas in the stratigraphic coordinates, this VGP is closest to the poles of the Mokulaevskaya (Mokulaevsky) and Kharaelakhskaya (Kharaelakhsky) formations for both calculation versions of the formation-mean poles (Figs. 6c, 6d, and 7; Table 3). Based on this, we may hypothesize that (1) the emplacement of the Siluriyskaya intrusion was

Table 2. Reference mean VGPs for volcanic formations of Norilsk region calculated from data of (Pavlov et al., 2019)

Formation	Lava flow mean				Mean over directional groups and individual directions			
	N	Plong°	Plat	A95°	*N*	Plong°	Plat	A95°
Ivakinskaya (Ivakinsky)	3	353.1	–51.1	16.5	2	350.7	–51.7	23.1
Syverminskaya (Syverminsky)	11	112.5	16.5	2.5	5	112.5	15.3	5.3
Gudchikhinskaya (Gudchikhinsky)	5	140.6	35.3	21.3	4	147.6	38.8	21.7
Tuklonskaya (Tuklonsky)	5	187.6	70.4	12.5	2	181.7	69.5	50.6
Nadezhdinskaya (Nadezhdinsky)	15	214.0	70.5	7.7	7	225.5	65.0	14.3
Morongovskaya (Morongovsky)	11	133.1	59.0	6.4	4	128.3	58.2	14.2
Mokulaevskaya (Mokulaevsky)	13	141.4	54.7	7.5	5	143.2	55.6	18.5
Kharaelakhskaya (Kharaelakhsky)	7	156.2	49.1	13.7	4	155.3	49.5	20.4
Kumginskaya (Kumginsky)	6	170.4	63.0	7.0	2	172.8	65.4	34.1

Table 3. Angular distances, ° between mean poles of intrusions in Kulumbe region and volcanic formations in Norilsk region. Formations: gd, Gudchikhinskaya (Gudchikhinsky); tk, Tuklonskaya (Tuklonsky); nd, Nadezhdinskaya (Nadezhdinsky); mr, Morongovskaya (Morongovsky); mk, Mokulaevskaya (Mokulaevsky); hr, Kharaelakhskaya (Kharaelakhsky); km, Kumginskaya (Kumginsky). CS, coordinate system: G, geographic, S, stratigraphic

Intrusions	CS	Formation flow-level mean VGPs							Formation pulse-level mean VGPs						
		gd	tk	nd	mr	mk	hr	km	gd	tk	nd	mr	mk	hr	km
Siluriyskaya	G	35.9	11.7	20.0	13.8	16.9	22.2	11.5	32.2	10.2	26.6	15.1	14.9	21.4	9.1
	S	15.5	25.7	34.4	12.4	5.7	4.2	16.1	10.1	22.5	39.3	13.9	4.3	3.5	17.8
Iltyk	G	30.1	35.1	42.9	13.7	18.9	29.4	31.1	31.0	33.0	49.5	11.1	18.6	28.3	30.8
	S	14.2	39.5	48.3	14.2	14.7	22.0	31.8	17.2	37.3	54.6	12.5	15.2	20.5	32.8
Kulyumbinskaya	G	49.7	18.3	20.7	25.5	30.2	36.8	24.0	46.6	18.6	27.2	26.1	29.4	35.9	21.0
	S	31.0	13.4	23.3	8.4	11.4	18.1	10.1	27.2	13.2	29.5	11.1	10.8	16.6	10.4
Kuzmovskii complex	G	41.6	12.0	18.0	18.4	22.1	27.8	15.2	38.0	11.4	25.1	19.7	21.1	27.3	13.3
	S	26.9	17.2	26.1	7.3	7.4	13.3	9.8	23.0	15.0	32.3	10.1	6.1	12.7	10.9
Khalil	G	15.5	33.9	42.6	9.3	8.3	16.9	25.9	16.0	31.7	48.8	8.3	9.8	16.2	27.4
	S	2.0	43.5	52.1	23.4	18.9	18.5	33.7	7.4	41.3	57.0	23.4	20.1	18.5	36.1
Katangskii complex	G	14.1	36.2	45.1	12.2	11.7	18.5	28.1	15.9	34.0	51.3	10.6	11.3	17.8	29.7
	S	4.7	44.9	53.2	27.6	22.2	18.5	34.8	6.3	42.7	57.6	27.9	23.0	18.5	37.4

probably close in time to the formation of the Mokulaevskaya (Mokulaevsky) and Kharaelakhskaya (Kharaelakhsky) lavas and (2) the rocks of the Siluriyskaya intrusion acquired the magnetization before tilting of the country rocks. We hypothesize a close age for the formation of the Norilsk-type ore-bearing intrusions located within the Norilsk trough—the Norilsk-1, Chernogorskii, and Norilsk-2 (Latyshev et al., 2020a).

Similar procedure was performed by us for some other intrusive bodies and groups of intrusions of the River Kulumbe. Thus, the Iltyk intrusion (site K8, the Kruglogorskii type of the Norilsk complex) demonstrates a pole that is significantly far from all reference poles in both coordinate systems and in both calculation options. Consequently, the Iltyk intrusion cannot be directly correlated with any of the formations in the Norilsk region according to paleomagnetic data. It should be emphasized that the results for the Iltyk intrusion were obtained for a single site, and more representative data are required for reliable correlation.

The mean pole over the sites of the Kulyumbinskaya intrusion in all calculation versions is distant from the poles of the volcanic formations of the Norilsk region. The smallest angular distance (8.4°, Table 3) is achieved in the comparison with the poles of the Morongovskaya (Morongovsky) Formation in calculation version I (Fig. 6c); therefore, as of now, this version of the correlation is, in our opinion, most preferable. The similar comparison results are obtained for the intrusive body attributed to the Kuz-

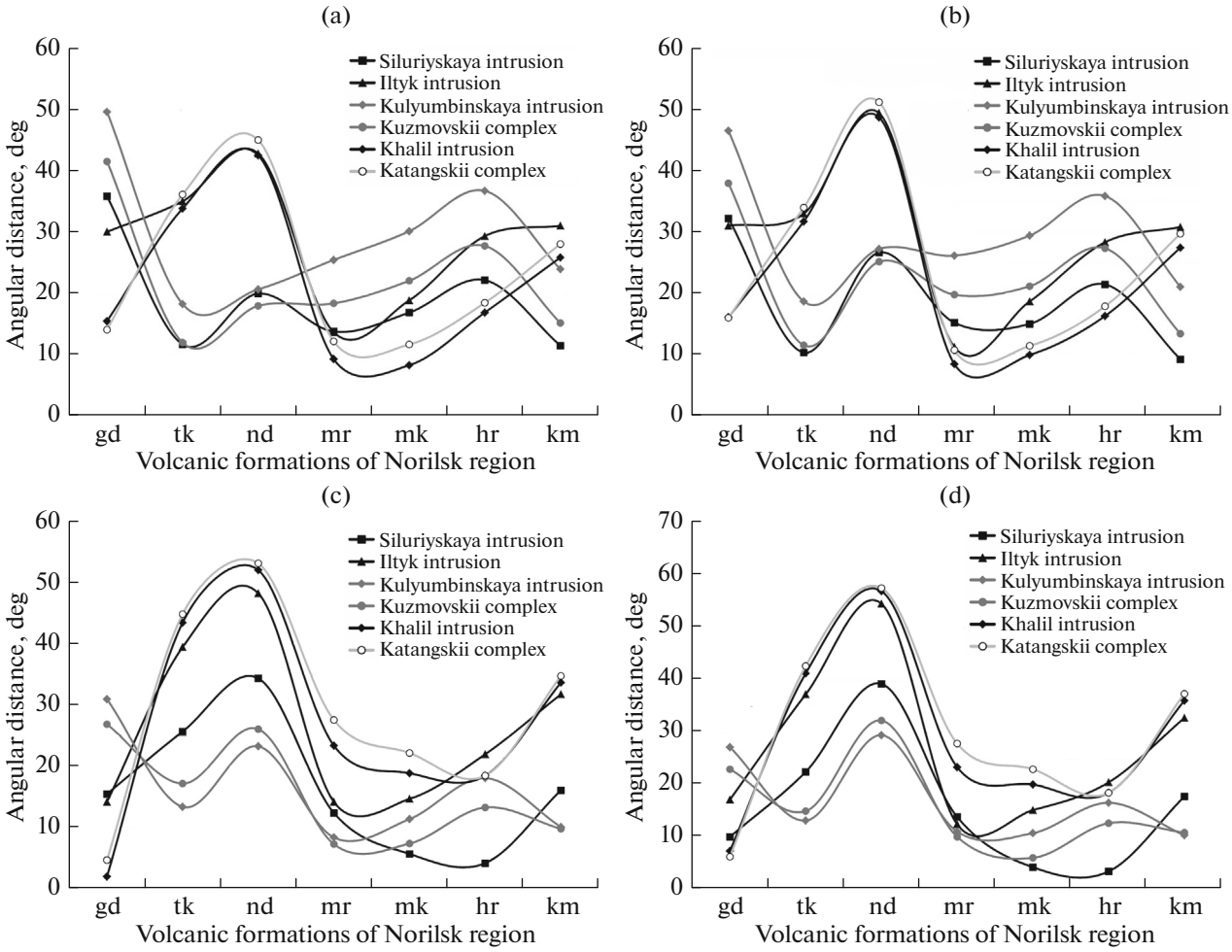


Fig. 6. Comparison of mean poles of studied intrusions and Norilsk volcanic formations: (a) intrusion poles in geographic coordinates; formation poles calculated on flow level; (b) intrusion poles in geographic coordinates; formation poles calculated on pulse level; (c) intrusion poles in stratigraphic coordinates; formation poles calculated on flow level; (d) intrusion poles in stratigraphic coordinates; formation poles calculated on pulse level. Formations: gd, Gudchikhinskaya (Gudchikhinsky), tk, Tuklonskaya (Tuklonsky); nd, Nadezhdinskaya (Nadezhdinsky); mr, Morongovskaya (Morongovsky); mk, Mokulaevskaya (Mokulaevsky); hr, Kharaelakhskaya (Kharaelakhsky); km, Kumginskaya (Kumginsky).

movskii complex (site K11) whose pole is close to that of the Kulyumbinskaya intrusion and can be correlated to the Morongovskaya (Morongovsky)–Mokulaevskaya (Mokulaevsky) formations (Figs. 6c, 6d). In Fig. 7 it can be seen that the poles of the Kulyumbinskaya intrusion and the bodies of the Kuzmovskii complex are located in the area between the poles of the Kumginskii, Morongovskii and Mokulaevskii formations, somewhat closer to the latter two in both versions of the curve of the reference poles. Thus, the correlation of the Kulyumbinskaya intrusion and the sill of the Kuzmovskii complex to the Morongovskaya (Morongovsky)–Mokulaevskaya (Mokulaevsky) formations seems to be most probable although the Kumginskaya (Kumginsky) age is also not completely excluded for these bodies. Besides, the closeness of the poles of the discussed intrusions indicates their approximately coeval formation despite the fact that these intrusions were referred to the different complexes (Kureyskii and Kuzmovskii). Moreover, it can be noted that the VGP of site KUL-40, also attributed to the Kureyskii complex (Krivolutskaya et al., 2020b), despite the large radius of the confidence circle, is close to the poles of the Kuzmovskii and Kulyumbinskaya intrusions and to the poles of the Morongovskaya (Morongovsky) Formation (Table 1). If the Kulyumbinskaya intrusion and other discussed bodies of the Kureyskii and Kuzmovskii complex are compared with the Morongovskaya (Morongovsky)–Mokulaevskaya (Mokulaevsky) formations, the emplacement time of these intrusions also proves to be close to the emplacement of the ore-bearing intrusions of the Norilsk region (Latyshev et al., 2020).

Considering the fact that the paleomagnetic directions of the Khalil intrusion significantly differ among the three sampled sites, for the comparison with the

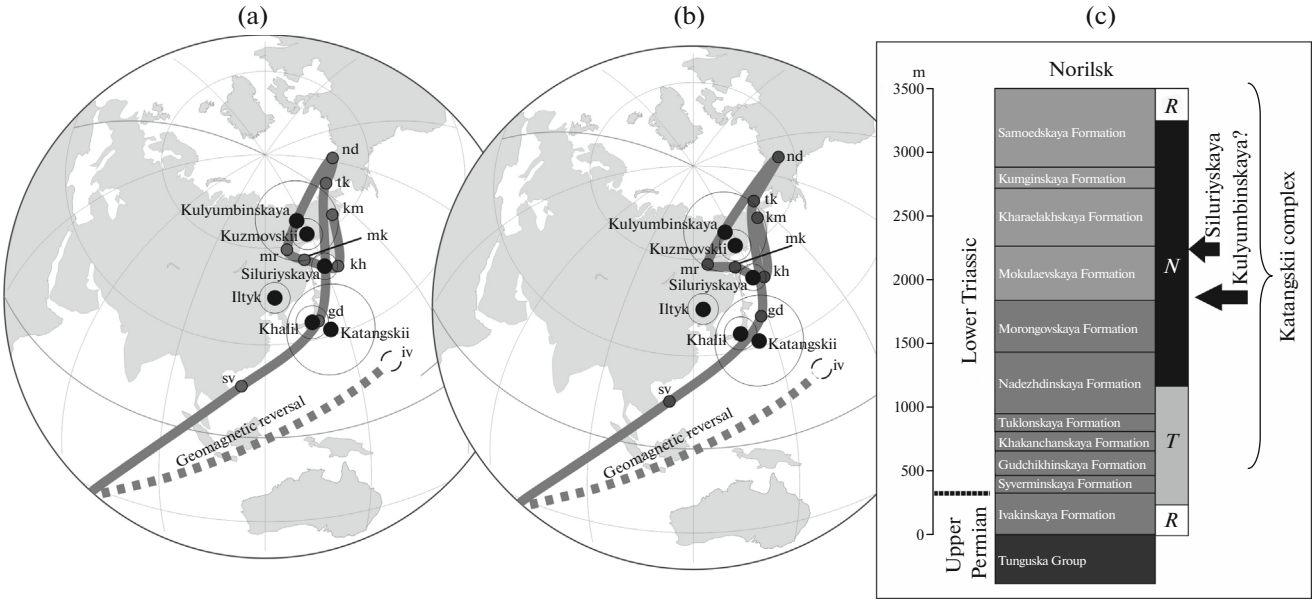


Fig. 7. Correlation of Kulumbe region intrusions with Norilsk tuff-lava sequence. (a), (b) Position of mean poles for studied intrusions (stratigraphic coordinate system) relative reference VGP curve over formations. Formation mean poles calculated (a) on flow level and (b) on level of directional groups and individual directions. Formations: iv, Ivakinskaya; sv, Syverminskaya; gd, Gudchikhskaya; tk, Tuklonskaya; nd, Nadezhdinskaya; mr, Morongovskaya; mk, Mokulaevskaya; hr, Kharaelakhskaya; km, Kumginskaya. (c) Correlation of Kulumbe region intrusions with Norilsk volcanic section. Magnetic stratigraphy of tuff-lava sequence is presented in (Heunemann et al., 2004; Latyshev et al., 2020b). *R* is interval of reversed polarity; *N* is interval of normal polarity; *T* is transition interval.

formations of the Norilsk region we used only the results for site K3 where the direction was calculated fairly accurately from quite a few samples. The only option to directly correlate the VGP of this site is to collate the pole in the stratigraphic coordinates with the Gudchikhinskaya (Gudchikhinsky) Formation (Figs. 6c, 6d, 7). Here, it should be taken into account that the Gudchikhinskii lavas were erupted during the geomagnetic reversal and that the mean paleomagnetic direction of this formation is the result of data averaging over several volcanic pulses and individual lava flows whose directions have a significant scatter (Gurevitch et al., 2004; Pavlov et al., 2019). Therefore, the correlation of the Khalil intrusion to the Norilsk tuff–lava sequence appears to be somewhat less reliable than for the previously discussed intrusions.

The mean VGP calculated from the intrusions of the Katangskii complex having normal polarity (without the "excursion" directions) corrected for bedding tilt of the country rock is also fairly close to the pole of the Gudchikhinskaya (Gudchikhinsky) Formation (Figs. 6c, 6d). Besides the above considerations, it should be taken into account that the individual VGPs over the sites of the Katangskii complex have a rather wide scatter ($A95_s = 14°$ for the mean) while the petrochemical characteristics of the Katangskii complex fundamentally differ from those of the Gudchikhinskaya (Gudchikhinsky) Formation and are, rather, close to the formations of the upper part of the Norilsk tuff-lava sequences (Krivolutskaya et al., 2020b). Given these facts, we prefer the interpretation according to which the formation of the Katangskii complex was distributed over time and spanned several stages of the deposition of the volcanic section. The presence of the above "excursion" directions at the individual sites indirectly supports the idea that the emplacement of the Katangskii intrusions could have begun during the transitional interval from the reversed to normal polarity encompassing the lower formations of the Norilsk volcanic section (Heunemann et al., 2004; Pavlov et al., 2019).

Thus, most of the poles calculated for the studied intrusions are quite far distant from the region of VGPs of the Norilsk volcanics in the geographic coordinate system while close to some formations in the stratigraphic coordinate system. This fact provides the additional evidence that the normally magnetized intrusions were formed before the end of regional deformations.

Finally, the mean VGP for the reversely magnetized intrusions are located more than 30° from the pole of the Ivakinskaya (Ivakinsky) Formation—the only reversely magnetized one in the Norilsk section. Taken together with our interpretation that at least part of the reversely magnetized bodies in the Kulumbe valley intruded after the end of main regional deformations, this testifies to the correlation of the reversely magnetized intrusions to the Daldykan complex corresponding to the final stage of the magmatism. We note that among the studied bodies there

are intrusions with petrochemical properties similar to the Katangskii complex, namely, the sites K5, K6, KUL-1, and KUL-25 (Krivolutskaya et al., 2020b). This indicates that the intrusive bodies of the Katangskii complex could be formed both during the epoch of normal polarity, synchronously with the volcanic activity, and during the subsequent epoch of the reversed polarity. The same time span for the formation of the intrusive bodies of the Katangskii complex was previously established in (Latyshev et al., 2018) for the central part of the Tunguska syncline. Thus, although the beginning time of the Katangskii complex formation remains unclear, the emplacement of Katangskii intrusions continued after the end of the main phase of the volcanism, approximately synchronously with the Daldykan complex in the Norilsk region.

Summarizing the discussion, we note that at different times, the intrusive magmatism in the Kulumbe valley region was localized at different stratigraphic levels. For example, the Carboniferous–Permian sediments of the Tunguska Group contain intrusive bodies having normal polarity (the Kulyumbinskaya intrusion) whereas the intrusions emplaced in the Cambrian–Ordovician formations are magnetized exclusively reversely (Fig. 1b). Only between these formations, in the Silurian–Devonian sediments both normally and reversely magnetized intrusions alternate, sometimes spaced within hundreds of meters apart from each other. However, in the latter case, the repeated manifestations of the magmatism can probably be associated with their location in the zone of the Imangda–Letninskiy fault which provides high permeability of the upper crustal horizons.

CONCLUSIONS

1. Based on the detailed analysis of the paleomagnetic data on the intrusive bodies in the Kulumbe river valley, we hypothesize that most of the studied intrusions were emplaced before the end of regional deformations which took place at the beginning of the Early Triassic at the final stages of trap magmatism. The reversely magnetized sills in the Turuka River mouth region are the exception: after their formation, significant deformations were absent.

2. The reversely magnetized intrusions of the Kulumbe valley region are most likely to have been formed after the end of the main phase of the volcanic activity in the Siberian platform, synchronously with the Daldykan stage of the magmatic activity in the Norilsk region.

3. The intrusions in the Kulumbe valley were formed for a fairly long time sufficient for averaging of the secular geomagnetic variation (10–100 kyr) and spanning at least one interval of normal and one interval of the reversed polarity at the Permian–Triassic boundary.

4. The Siluriyskaya intrusion (Norilsk type) by its age can be correlated to the Mokulaevskaya–Kharaelakhskaya formations of the Norilsk region (Fig. 7c). This indicates that the formation of the intrusions of the Norilsk type was close in time to sulfide mineralization in the Norilsk and Kulumbe regions.

5. The Kulyumbinskaya intrusion and some other bodies referred to the Kureiskii and Kuzmovkskii complexes are most likely to correspond to the Morongovskaya–Mokulaevskaya time.

6. The Katangskii complex of the intrusions was formed for a fairly long time synchronously with the main phase of the volcanic activity in the Siberian Platform and perhaps after its cessation. Thus, the Katangskii complex cannot be directly correlated to any specific formation of the Norilsk region.

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